

# Challenge: Bike Count Estimation

## Supervised Learning

Jorunn Mense & Karsten Schroedter & Fabian Gieseke

Machine Learning and Data Engineering  
Department of Information Systems  
University of Münster

# Outline

1 Motivation

2 Your Task

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## 1 Motivation

## 2 Your Task

## Münster: bicycle city

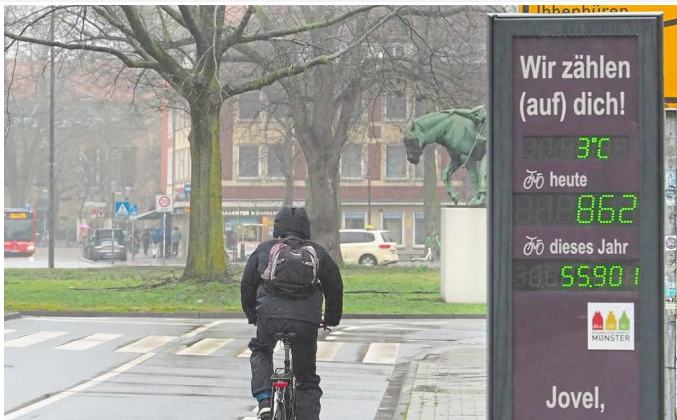


## Münster: bicycle city

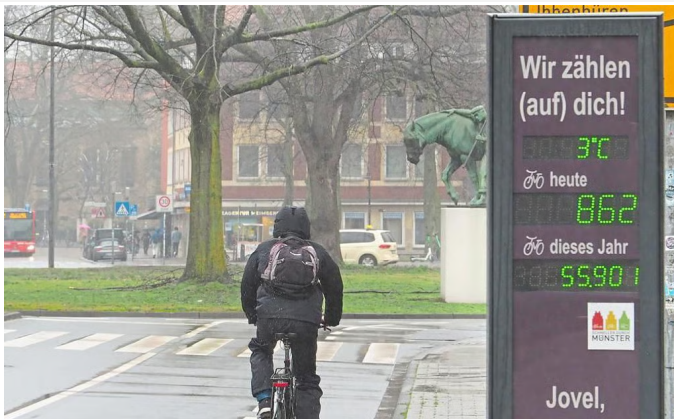


But how many are there?

# Counting Bikes

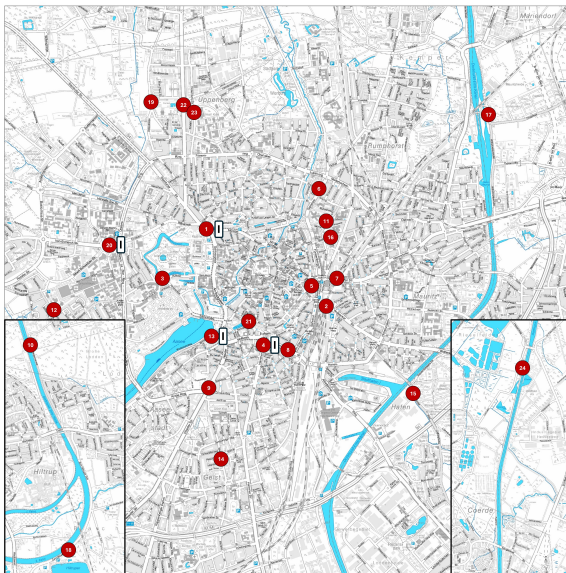


## Counting Bikes



The city does count the bikes!

## Counting at multiple locations



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## Dataset

| Month | Day | Hour | Weekday | Weather        | Temperature (°C) | Humidity (%) | Rain (mm) | Wind (km/h) | BikeCount |
|-------|-----|------|---------|----------------|------------------|--------------|-----------|-------------|-----------|
| 1     | 1   | 0    | 6       | Sonnig         | 14               | 61           | 0         | 34          | 73        |
| 1     | 1   | 1    | 6       | Sonnig         | 14               | 59           | 0         | 34          | 193       |
| 1     | 1   | 2    | 6       | Leicht bewölkt | 14               | 57           | 0         | 33          | 240       |
| 1     | 1   | 3    | 6       | Leicht bewölkt | 14               | 55           | 0         | 33          | 279       |
| 1     | 1   | 4    | 6       | Leicht bewölkt | 14               | 56           | 0         | 33          | 194       |
| 1     | 1   | 5    | 6       | Leicht bewölkt | 14               | 57           | 0         | 33          | 112       |
| 1     | 1   | 6    | 6       | Leicht bewölkt | 14               | 58           | 0         | 33          | 45        |
| 1     | 1   | 7    | 6       | Leicht bewölkt | 14               | 61           | 0         | 31          | 21        |
| 1     | 1   | 8    | 6       | Leicht bewölkt | 13               | 64           | 0         | 29          | 26        |
| 1     | 1   | 9    | 6       | Leicht bewölkt | 13               | 66           | 0         | 26          | 24        |
| 1     | 1   | 10   | 6       | Leicht bewölkt | 13               | 66           | 0         | 28          | 26        |
| 1     | 1   | 11   | 6       | Leicht bewölkt | 14               | 65           | 0         | 29          | 55        |
| 1     | 1   | 12   | 6       | Leicht bewölkt | 14               | 65           | 0         | 31          | 115       |
| 1     | 1   | 13   | 6       | Leicht bewölkt | 14               | 67           | 0         | 29          | 116       |
| 1     | 1   | 14   | 6       | Bedeckt        | 13               | 70           | 0         | 27          | 165       |
| 1     | 1   | 15   | 6       | Bedeckt        | 13               | 73           | 0         | 26          | 183       |
| 1     | 1   | 16   | 6       | Bedeckt        | 12               | 75           | 0         | 23          | 163       |
| 1     | 1   | 17   | 6       | Bedeckt        | 12               | 77           | 0         | 19          | 153       |
| 1     | 1   | 18   | 6       | Bedeckt        | 12               | 80           | 0         | 16          | 118       |
| 1     | 1   | 19   | 6       | Bedeckt        | 12               | 79           | 0         | 16          | 107       |
| 1     | 1   | 20   | 6       | Bedeckt        | 12               | 79           | 0         | 17          | 93        |
| 1     | 1   | 21   | 6       | Bedeckt        | 12               | 79           | 0         | 18          | 57        |
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You will get one year of hourly data.

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## Your Task

- Develop a machine learning model that predicts the bike count of the next hours.

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- Try out different models from the lecture and compare their performance  $\Rightarrow$  **at least** try out one linear model, one tree-based model and one neural network.
- Compare performance for two forecast horizons:
  - 1 predict bike count of **the next hour**
  - 2 predict bike count for **the next 24 hours**

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- Compare performance for two forecast horizons:
  - 1 predict bike count of **the next hour**
  - 2 predict bike count for **the next 24 hours**
- **Key aspects:** This is a time series dataset. Think about how to use this, e.g. in feature engineering!

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Also add the MSE of the hidden test set to your presentation!

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- You will present your two slides in a 3 minute talk on 3rd of June. Everyone from the group should present one part!
- You will have to submit your presentation and Jupyter Notebook on **2nd of June** via Learnweb.